Data Privacy: Homeworks #4 Solutions

Due on Wednesday 11PM, June 17, 2026

Note: Submit homework via LearnUS. Topics: membership inference attacks and machine unlearning. Throughout this homework, you may assume that all random variables you encounter have continuous densities on $\mathbb{R}$ (or $\mathbb{R}^d$) and that all integrals are well-defined. No measure theory is required.

1. **Neyman–Pearson Lemma.** Let T be a real-valued random variable. Under hypothesis H_b (for $b \in \{0,1\}$), assume T has a continuous density p_b on $\mathbb{R}$, with $p_0(t) > 0$ for every t. Define the *likelihood ratio*

$$\Lambda(t) = \frac{p_1(t)}{p_0(t)}.$$

A *test* is a function $\phi : \mathbb{R} \to [0,1]$, where $\phi(t)$ is the probability of rejecting H_0 after observing $T = t$. Define

$$\alpha(\phi) = \mathbb{E}_{H_0}[\phi(T)] \quad \text{(size, type-I error)}, \qquad \beta(\phi) = \mathbb{E}_{H_1}[\phi(T)] \quad \text{(power)}.$$

Fix a level $\alpha_0 \in (0,1)$, and *assume that the function* $G(\tau) := \Pr_{H_0}[\Lambda(T) > \tau]$ *is continuous on* $[0,\infty)$. (This holds automatically in the Gaussian models you will see in Problems 3, 5, and 6.) Prove the following three claims.

(a) **Existence.** Show that there exists a threshold $\tau^* \geq 0$ such that the test

$$\phi^*(t) = \mathbf{1}[\Lambda(t) > \tau^*]$$

satisfies $\alpha(\phi^*) = \alpha_0$ exactly.
Hint: Show that G is non-increasing on $[0,\infty)$ with $G(0) \leq 1$ and $G(\tau) \to 0$ as $\tau \to \infty$. Then apply the intermediate value theorem.

(b) **Optimality.** Show that for every test ϕ with $\alpha(\phi) \leq \alpha_0$, we have $\beta(\phi) \leq \beta(\phi^*)$.
Hint: Verify that for every t, $(\phi^*(t) - \phi(t))\,(p_1(t) - \tau^*\, p_0(t)) \geq 0$, then integrate over $\mathbb{R}$.

(c) **Uniqueness.** Suppose ϕ is any most-powerful level-α_0 test (i.e., $\alpha(\phi) \leq \alpha_0$ and $\beta(\phi) = \beta(\phi^*)$). Show that $\phi(t) = \phi^*(t)$ for every t with $\Lambda(t) \neq \tau^*$, except possibly on a set of probability 0.
Hint: The integrand from part (b) is non-negative and now integrates to 0.

Solution:

(a) **Existence of τ^*.** If $\tau_1 \leq \tau_2$ then $\{\Lambda > \tau_2\} \subseteq \{\Lambda > \tau_1\}$, so $G(\tau_2) \leq G(\tau_1)$. Hence G is non-increasing on $[0,\infty)$. Also,

$$G(0) = \Pr_{H_0}[\Lambda(T) > 0] \leq 1, \qquad G(\tau) \xrightarrow{\tau\to\infty} 0,$$

because $\{\Lambda > \tau\}$ shrinks to the empty event as $\tau \to \infty$. By the assumed continuity of G and the intermediate value theorem, there exists $\tau^* \geq 0$ with $G(\tau^*) = \alpha_0$. Then

$$\alpha(\phi^*) = \mathbb{E}_{H_0}[\mathbf{1}[\Lambda(T) > \tau^*]] = \Pr_{H_0}[\Lambda(T) > \tau^*] = G(\tau^*) = \alpha_0.$$

(b) **Optimality.** For each fixed t we check the pointwise inequality $(\phi^*(t) - \phi(t))(p_1(t) - \tau^* p_0(t)) \geq 0$ case by case:

- If $p_1(t) > \tau^* p_0(t)$: $\phi^*(t) = 1 \geq \phi(t)$ and the second factor is > 0, so the product is ≥ 0.
- If $p_1(t) < \tau^* p_0(t)$: $\phi^*(t) = 0 \leq \phi(t)$ and the second factor is < 0, so the product is ≥ 0.
- If $p_1(t) = \tau^* p_0(t)$: the second factor is 0.

Integrate the inequality over $\mathbb{R}$:

$$0 \leq \int (\phi^*(t) - \phi(t))(p_1(t) - \tau^* p_0(t))\, dt = (\beta(\phi^*) - \beta(\phi)) - \tau^*(\alpha(\phi^*) - \alpha(\phi)).$$

Since $\alpha(\phi^*) = \alpha_0 \geq \alpha(\phi)$ and $\tau^* \geq 0$, the second term on the right is non-negative, so $\beta(\phi^*) - \beta(\phi) \geq \tau^*(\alpha_0 - \alpha(\phi)) \geq 0$.

(c) **Uniqueness.** If ϕ is also most powerful, $\beta(\phi) = \beta(\phi^*)$, and combining with $\alpha(\phi) \leq \alpha_0 = \alpha(\phi^*)$,

$$0 \leq \int (\phi^* - \phi)(p_1 - \tau^* p_0)\, dt = -\tau^*(\alpha_0 - \alpha(\phi)) \leq 0,$$

so the integral equals 0. The integrand is ≥ 0 everywhere, and its integral is 0, so it must equal 0 except possibly on a set of probability 0. On $\{t : p_1(t) > \tau^* p_0(t)\}$ the second factor is strictly positive, forcing $\phi^*(t) - \phi(t) = 0$ there (except on a probability-zero set); since $\phi^* = 1$ on this set, $\phi = 1$ there. Symmetrically, $\phi = 0 = \phi^*$ on $\{p_1 < \tau^* p_0\}$. Hence ϕ agrees with ϕ^* on $\{\Lambda \neq \tau^*\}$ (apart from a probability-zero set).

2. **DP $\Rightarrow$ MIA Bounds.** Let $M : \mathcal{X}^n \to \mathcal{Y}$ be (ε, δ)-differentially private, and let D_0, D_1 be two neighboring datasets. Consider any (possibly randomized) binary test $\mathcal{A} : \mathcal{Y} \to \{0, 1\}$ that tries to distinguish $M(D_0)$ from $M(D_1)$. Write

$$\mathrm{FPR}(\mathcal{A}) = \Pr[\mathcal{A}(M(D_0)) = 1], \qquad \mathrm{TPR}(\mathcal{A}) = \Pr[\mathcal{A}(M(D_1)) = 1].$$

(a) Show that

$$\mathrm{TPR}(\mathcal{A}) \leq e^{\varepsilon}\, \mathrm{FPR}(\mathcal{A}) + \delta, \qquad 1 - \mathrm{FPR}(\mathcal{A}) \leq e^{\varepsilon}\, (1 - \mathrm{TPR}(\mathcal{A})) + \delta.$$

(b) Define the MI advantage $\mathrm{Adv}(\mathcal{A}) = \mathrm{TPR}(\mathcal{A}) - \mathrm{FPR}(\mathcal{A})$. Show the loose bound

$$\mathrm{Adv}(\mathcal{A}) \leq e^{\varepsilon} - 1 + \delta.$$

Then, by combining both inequalities of part (a), derive the tighter Kairouz–Oh–Viswanath bound

$$\mathrm{Adv}(\mathcal{A}) \leq \frac{e^{\varepsilon} - 1 + 2\delta}{e^{\varepsilon} + 1}.$$

(c) Conclude that for $\varepsilon = 0.1$ and $\delta = 10^{-5}$, no adversary can exceed advantage ≈ 0.105, but for $\varepsilon = 8$ the bound is vacuous.

Solution:

(a) Let $S = \{y : \mathcal{A}(y) = 1\} \subseteq \mathcal{Y}$ (if $\mathcal{A}$ is randomized, average over its internal randomness; this does not affect the DP inequality). Apply the (ε, δ)-DP guarantee to S with the ordered pair (D_1, D_0):

$$\Pr[M(D_1) \in S] \le e^{\varepsilon} \Pr[M(D_0) \in S] + \delta \iff \mathrm{TPR} \le e^{\varepsilon}\mathrm{FPR} + \delta.$$

Apply it to the complementary event S^c with the ordered pair (D_0, D_1):

$$\Pr[M(D_0) \in S^c] \le e^{\varepsilon} \Pr[M(D_1) \in S^c] + \delta \iff 1 - \mathrm{FPR} \le e^{\varepsilon}(1 - \mathrm{TPR}) + \delta.$$

(b) **Loose bound.** From the first inequality and $\mathrm{FPR} \le 1$,

$$\mathrm{Adv} = \mathrm{TPR} - \mathrm{FPR} \le (e^{\varepsilon} - 1)\mathrm{FPR} + \delta \le e^{\varepsilon} - 1 + \delta.$$

Tighter (Kairouz–Oh–Viswanath). The pair of inequalities from (a) defines a polygonal region in the $(\mathrm{FPR}, \mathrm{TPR})$-plane. The maximum of $\mathrm{TPR} - \mathrm{FPR}$ over this region is attained where both inequalities are tight. Rewriting the second inequality as

$$\mathrm{TPR} \ge 1 - e^{-\varepsilon}(1 - \mathrm{FPR}) - \delta e^{-\varepsilon},$$

and setting equal to $e^{\varepsilon}\mathrm{FPR} + \delta$, we get

$$(e^{\varepsilon} - e^{-\varepsilon})\mathrm{FPR} = (1 - e^{-\varepsilon})(1 - \delta) \implies \mathrm{FPR}^{\star} = \frac{1 - \delta}{e^{\varepsilon} + 1}.$$

Then $\mathrm{TPR}^{\star} = e^{\varepsilon}\mathrm{FPR}^{\star} + \delta = (e^{\varepsilon} + \delta)/(e^{\varepsilon} + 1)$, so

$$\mathrm{Adv}^{\star} = \mathrm{TPR}^{\star} - \mathrm{FPR}^{\star} = \frac{e^{\varepsilon} + \delta - (1 - \delta)}{e^{\varepsilon} + 1} = \frac{e^{\varepsilon} - 1 + 2\delta}{e^{\varepsilon} + 1}.$$

(c) For $\varepsilon = 0.1, \delta = 10^{-5}$: loose bound $= e^{0.1} - 1 + 10^{-5} \approx 0.1052$; tighter bound ≈ 0.0500. For $\varepsilon = 8$: $e^8 \approx 2981$, so the loose bound is ≈ 2980 and the tighter bound is ≈ 1. Both are useless — advantage is automatically ≤ 1.

3. **Bayes-Optimal Membership Test.** Let $B \in \{0, 1\}$ be a Bernoulli(π) membership indicator, and let T be an observed statistic with continuous density $p_b(t)$ given $B = b$. Consider any decision rule $\hat{B} : \mathbb{R} \to \{0, 1\}$, and measure its quality by the 0-1 error $\Pr[\hat{B}(T) \neq B]$.

   (a) Compute the posterior $\Pr[B = 1 \mid T = t]$ in terms of p_0, p_1, π.

   (b) Show that the Bayes-optimal rule (the MAP rule) is

   $$\hat{B}_{\mathrm{MAP}}(t) = \mathbf{1}\left[\Lambda(t) \ge \frac{1 - \pi}{\pi}\right], \qquad \Lambda(t) = \frac{p_1(t)}{p_0(t)},$$

   and connect this to Problem 1: the Bayes-optimal MIA is a likelihood-ratio test, and only the threshold changes with the prior.

(c) Suppose $T = \ell(f_\theta, z)$ is the loss of model f_θ on the queried point z, and that $\ell \mid B = 1 \sim \mathcal{N}(\mu_{\text{in}}, \sigma^2)$, $\ell \mid B = 0 \sim \mathcal{N}(\mu_{\text{out}}, \sigma^2)$ with $\mu_{\text{in}} < \mu_{\text{out}}$. Show that the optimal rule reduces to a single-threshold loss test $\hat{B} = \mathbf{1}[\ell \le c]$ and compute c explicitly when $\pi = 1/2$.

Solution:

(a) By Bayes' rule,

$$\Pr[B = 1 \mid T = t] = \frac{\pi\, p_1(t)}{\pi\, p_1(t) + (1 - \pi)\, p_0(t)}.$$

(b) The Bayes (0-1 loss) rule outputs $\hat{B} = 1$ iff $\Pr[B = 1 \mid T = t] \ge \Pr[B = 0 \mid T = t]$, i.e.,

$$\pi\, p_1(t) \ge (1 - \pi)\, p_0(t) \iff \Lambda(t) := \frac{p_1(t)}{p_0(t)} \ge \frac{1 - \pi}{\pi}.$$

By Problem 1, this is exactly a Neyman–Pearson likelihood-ratio test with threshold $\tau = (1 - \pi)/\pi$. Changing the prior π only changes the threshold; the *form* of the optimal MIA is always a likelihood-ratio test.

(c) **Gaussian case.** With shared variance σ^2,

$$\log \Lambda(\ell) = -\frac{(\ell - \mu_{\text{in}})^2 - (\ell - \mu_{\text{out}})^2}{2\sigma^2} = \frac{(\mu_{\text{out}} - \mu_{\text{in}})(\mu_{\text{in}} + \mu_{\text{out}} - 2\ell)}{2\sigma^2},$$

a strictly decreasing affine function of ℓ (since $\mu_{\text{out}} > \mu_{\text{in}}$). Hence $\log \Lambda(\ell) \ge \log((1 - \pi)/\pi)$ iff

$$\ell \le c = \frac{\mu_{\text{in}} + \mu_{\text{out}}}{2} - \frac{\sigma^2}{\mu_{\text{out}} - \mu_{\text{in}}} \log \frac{1 - \pi}{\pi}.$$

For $\pi = 1/2$ the log term vanishes, giving $c = (\mu_{\text{in}} + \mu_{\text{out}})/2$.

4. **Generalization Gap as a Leakage Lower Bound (Yeom).** Suppose the loss is bounded: $\ell(f_\theta, z) \in [0, B]$ for all θ, z. Define

$$R_{\text{train}} = \mathbb{E}_{z \sim \text{Unif}(D)}[\ell(f_\theta, z)], \qquad R_{\text{pop}} = \mathbb{E}_{z \sim \mathcal{P}}[\ell(f_\theta, z)], \qquad \Delta = R_{\text{pop}} - R_{\text{train}} \ge 0.$$

The MIA game: sample $b \sim \text{Unif}\{0, 1\}$. If $b = 1$, return z uniform over D; otherwise, return $z \sim \mathcal{P}$. The adversary uses a threshold attack $\mathcal{A}_\tau(z) = \mathbf{1}[\ell(f_\theta, z) \le \tau]$. Let $F_1(\tau) = \Pr[\ell \le \tau \mid b = 1]$ and $F_0(\tau) = \Pr[\ell \le \tau \mid b = 0]$; assume both have continuous densities on $[0, B]$.

(a) Show that the advantage of $\mathcal{A}_\tau$ is

$$\text{Adv}(\mathcal{A}_\tau) = F_1(\tau) - F_0(\tau).$$

(b) Prove the *layer-cake identity*: for any random variable X with continuous density on $[0, B]$,

$$\mathbb{E}[X] = \int_0^B (1 - F_X(t))\, dt.$$

Then deduce

$$\int_0^B (F_1(t) - F_0(t))\, dt = \Delta.$$

(c) Conclude that there exists a threshold $\tau^\star \in [0, B]$ for which the threshold attack achieves advantage

$$\mathrm{Adv}(\mathcal{A}_{\tau^\star}) \geq \frac{\Delta}{B}.$$

In words: the generalization gap is a *lower* bound on the leakage of the best threshold attack — not an upper bound, contrary to a common informal slogan.
Hint: $\Delta = \int_0^B (F_1 - F_0)\, dt \leq B \cdot \sup_t (F_1(t) - F_0(t))$.

(d) Now suppose additionally that under each hypothesis the loss is Gaussian:

$$\ell \mid b = 1 \sim \mathcal{N}(R_{\text{train}}, \sigma^2), \qquad \ell \mid b = 0 \sim \mathcal{N}(R_{\text{pop}}, \sigma^2).$$

(Ignore the issue that Gaussians are not literally bounded; the formula below is the correct one for the full range $\mathbb{R}$.) Show that the maximum-advantage threshold attack has $\tau^\star = (R_{\text{train}} + R_{\text{pop}})/2$ and achieves

$$\mathrm{Adv}^\star = 2\Phi\left(\frac{\Delta}{2\sigma}\right) - 1.$$

For small Δ/σ, derive the linear approximation $\mathrm{Adv}^\star \approx \Delta/(\sigma\sqrt{2\pi})$ and explain why noisier loss distributions reduce leakage.

Solution:

(a) By definitions, $\mathrm{TPR} = \Pr[\mathcal{A}_\tau = 1 \mid b = 1] = \Pr[\ell \leq \tau \mid b = 1] = F_1(\tau)$ and $\mathrm{FPR} = F_0(\tau)$. Subtracting gives $\mathrm{Adv}(\mathcal{A}_\tau) = F_1(\tau) - F_0(\tau)$.

(b) **Layer cake.** Let f_X be the density of X. Write $t = \int_0^t 1\, ds$ and substitute:

$$\mathbb{E}[X] = \int_0^B t f_X(t)\, dt = \int_0^B \left(\int_0^t 1\, ds\right) f_X(t)\, dt = \iint_{0 \leq s < t \leq B} f_X(t)\, ds\, dt.$$

Switch the order of integration over the triangle $\{0 \leq s < t \leq B\} = \{0 \leq s \leq B,\ s < t \leq B\}$:

$$\mathbb{E}[X] = \int_0^B \left(\int_s^B f_X(t)\, dt\right) ds = \int_0^B \Pr[X > s]\, ds = \int_0^B (1 - F_X(s))\, ds.$$

Apply this to ℓ under each hypothesis:

$$R_{\text{pop}} - R_{\text{train}} = \int_0^B [(1 - F_0(t)) - (1 - F_1(t))]\, dt = \int_0^B (F_1(t) - F_0(t))\, dt = \Delta.$$

(c) Let $h(t) = F_1(t) - F_0(t)$. If $h(t) < \Delta/B$ for every $t \in [0, B]$, then $\int_0^B h\,dt < B \cdot (\Delta/B) = \Delta$, contradicting (b). Hence some $\tau^\star \in [0, B]$ satisfies $h(\tau^\star) \geq \Delta/B$, so $\mathrm{Adv}(\mathcal{A}_{\tau^\star}) \geq \Delta/B$. The generalization gap is therefore a *lower* bound on the leakage of the best threshold attack, not an upper bound.

(d) **Gaussian.** With $\mu_1 = R_{\text{train}}, \mu_0 = R_{\text{pop}}$ we have $F_b(t) = \Phi((t - \mu_b)/\sigma)$, so

$$h(\tau) = \Phi\left(\frac{\tau - R_{\text{train}}}{\sigma}\right) - \Phi\left(\frac{\tau - R_{\text{pop}}}{\sigma}\right).$$

Setting $h'(\tau) = 0$ gives $\phi((\tau - R_{\text{train}})/\sigma) = \phi((\tau - R_{\text{pop}})/\sigma)$ (where ϕ is the standard normal pdf), so $|\tau - R_{\text{train}}| = |\tau - R_{\text{pop}}|$, i.e., $\tau^\star = (R_{\text{train}} + R_{\text{pop}})/2$. Then $\tau^\star - R_{\text{train}} = \Delta/2$ and $\tau^\star - R_{\text{pop}} = -\Delta/2$, so

$$\mathrm{Adv}^\star = \Phi(\Delta/(2\sigma)) - \Phi(-\Delta/(2\sigma)) = 2\Phi(\Delta/(2\sigma)) - 1.$$

For small $x = \Delta/(2\sigma)$, $\Phi(x) \approx 1/2 + x\phi(0) = 1/2 + x/\sqrt{2\pi}$, so $\mathrm{Adv}^\star \approx 2x/\sqrt{2\pi} = \Delta/(\sigma\sqrt{2\pi})$. Larger σ (noisier loss distributions) flattens the gap and reduces leakage — the same idea behind DP noise injection.

5. **LiRA Closed Form and Decision Boundary.** Assume that for a target record z, the per-shadow-model loss has Gaussian distributions under the two membership hypotheses:

$$\ell \mid (z \in D) \sim \mathcal{N}(\mu_{\text{in}}, \sigma_{\text{in}}^2), \qquad \ell \mid (z \notin D) \sim \mathcal{N}(\mu_{\text{out}}, \sigma_{\text{out}}^2).$$

(a) Derive the log-likelihood ratio

$$\log \Lambda(\ell) = \log \frac{\mathcal{N}(\ell; \mu_{\text{in}}, \sigma_{\text{in}}^2)}{\mathcal{N}(\ell; \mu_{\text{out}}, \sigma_{\text{out}}^2)} = \frac{(\ell - \mu_{\text{out}})^2}{2\sigma_{\text{out}}^2} - \frac{(\ell - \mu_{\text{in}})^2}{2\sigma_{\text{in}}^2} + \log \frac{\sigma_{\text{out}}}{\sigma_{\text{in}}}.$$

(b) Show that for general $\sigma_{\text{in}} \neq \sigma_{\text{out}}$, the optimal decision region $\{\log \Lambda > \log \tau\}$ is one of: a bounded interval, the complement of a bounded interval, a half-line, all of $\mathbb{R}$, or empty. Identify which case occurs based on the sign of $\sigma_{\text{in}} - \sigma_{\text{out}}$.

(c) Show that in the homoscedastic case $\sigma_{\text{in}} = \sigma_{\text{out}} = \sigma$, the optimal MIA reduces to the single threshold $\mathcal{A}(z) = \mathbf{1}[\ell \leq c]$, and express c in terms of $\mu_{\text{in}}, \mu_{\text{out}}, \sigma, \tau$.

(d) In the homoscedastic case, derive the closed-form ROC: at false-positive rate α, the optimal true-positive rate is

$$\mathrm{TPR}(\alpha) = \Phi\left(\Phi^{-1}(\alpha) + \frac{\mu_{\text{out}} - \mu_{\text{in}}}{\sigma}\right),$$

where Φ is the standard normal CDF.

Solution:

(a) The Gaussian log-density is $\log \mathcal{N}(\ell; \mu, \sigma^2) = -\frac{(\ell-\mu)^2}{2\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2)$. Subtracting,

$$\log \Lambda(\ell) = \frac{(\ell - \mu_{\text{out}})^2}{2\sigma_{\text{out}}^2} - \frac{(\ell - \mu_{\text{in}})^2}{2\sigma_{\text{in}}^2} + \log \frac{\sigma_{\text{out}}}{\sigma_{\text{in}}}.$$

(b) Expanding, $\log \Lambda(\ell)$ is a quadratic in ℓ with leading coefficient

$$A = \frac{1}{2}\left(\frac{1}{\sigma_{\text{out}}^2} - \frac{1}{\sigma_{\text{in}}^2}\right).$$

- $\sigma_{\text{in}} < \sigma_{\text{out}}$ (members are more concentrated): $A < 0$, so $\log \Lambda$ is concave in ℓ. Then $\{\log \Lambda > \log \tau\}$ is a (possibly empty, possibly unbounded) *interval* (c_-, c_+) on the loss axis.
- $\sigma_{\text{in}} > \sigma_{\text{out}}$: $A > 0$, so $\log \Lambda$ is convex. Then $\{\log \Lambda > \log \tau\}$ is the *complement of a bounded interval*: $\{\ell < c_-\} \cup \{\ell > c_+\}$.
- $\sigma_{\text{in}} = \sigma_{\text{out}}$: $A = 0$, $\log \Lambda$ is linear, and the region is a half-line.

(c) **Homoscedastic.** With $\sigma_{\text{in}} = \sigma_{\text{out}} = \sigma$ the quadratic terms cancel:

$$\log \Lambda(\ell) = \frac{(\ell - \mu_{\text{out}})^2 - (\ell - \mu_{\text{in}})^2}{2\sigma^2} = \frac{(\mu_{\text{out}} - \mu_{\text{in}})(\mu_{\text{in}} + \mu_{\text{out}} - 2\ell)}{2\sigma^2},$$

a strictly decreasing affine function of ℓ (since $\mu_{\text{out}} > \mu_{\text{in}}$). Hence $\log \Lambda \geq \log \tau$ iff $\ell \leq c$, where

$$c = \frac{\mu_{\text{in}} + \mu_{\text{out}}}{2} - \frac{\sigma^2 \log \tau}{\mu_{\text{out}} - \mu_{\text{in}}}.$$

(d) $\text{FPR}(c) = \Pr_{\text{out}}[\ell \leq c] = \Phi((c - \mu_{\text{out}})/\sigma)$. Solving for c at $\text{FPR} = \alpha$: $c = \mu_{\text{out}} + \sigma\Phi^{-1}(\alpha)$. Substituting,

$$\text{TPR}(\alpha) = \Pr_{\text{in}}[\ell \leq c] = \Phi\left(\frac{c - \mu_{\text{in}}}{\sigma}\right) = \Phi\left(\Phi^{-1}(\alpha) + \frac{\mu_{\text{out}} - \mu_{\text{in}}}{\sigma}\right).$$

The discrimination is controlled entirely by the standardized gap $(\mu_{\text{out}} - \mu_{\text{in}})/\sigma$.

6. **Concrete MIA on a Gaussian Mean Model.** A "training" algorithm maps $D = (z_1, \ldots, z_n)$ with $z_i \in \mathbb{R}$ to the parameter

$$\theta(D) = \bar{z} + W, \quad \bar{z} = \frac{1}{n}\sum_{i=1}^{n} z_i, \quad W \sim \mathcal{N}(0, \rho^2) \text{ independent of } D.$$

Assume the population $\mathcal{P}$ is $\mathcal{N}(0, 1)$, and the adversary observes θ plus a candidate point z^*. The MIA game: $b \sim \text{Unif}\{0, 1\}$. If $b = 1$, $z^* = z_1$ (a training sample, included in D); if $b = 0$, $z^* \sim \mathcal{P}$ independent of D.

(a) Show that conditional on z^*,

$$\theta \mid (b = 1) \sim \mathcal{N}\left(\frac{z^*}{n}, \ \frac{n-1}{n^2} + \rho^2\right), \qquad \theta \mid (b = 0) \sim \mathcal{N}\left(0, \ \frac{1}{n} + \rho^2\right).$$

(b) Show that the optimal adversary's test statistic is the (conditional) likelihood ratio $\Lambda(\theta, z^*) = p(\theta \mid b = 1, z^*)/p(\theta \mid b = 0, z^*)$, and write down $\log \Lambda$ as a quadratic in θ (with z^* treated as a known constant).

(c) Now suppose the inputs are clipped, $|z_i| \le 1$, so the L_2-sensitivity of $\bar{z}$ under one-element neighbors is at most $2/n$. The Gaussian mechanism with

$$\rho \ge \frac{2}{n\varepsilon}\sqrt{2\log(1.25/\delta)}$$

makes the release (ε, δ)-DP. Using Problem 2(b), give a numerical upper bound on the optimal MIA advantage for $\varepsilon = 0.1$, $\delta = 10^{-5}$, and comment on why $\varepsilon = 1$ already gives a vacuous bound.

Solution:

(a) **Under $b = 0$:** z^* is drawn fresh from $\mathcal{P}$, independent of D. Since $\bar{z} = (1/n)\sum_{i=1}^{n} z_i$ is the average of n i.i.d. $\mathcal{N}(0,1)$ random variables, $\bar{z} \sim \mathcal{N}(0, 1/n)$. Adding the independent Gaussian noise, $\theta = \bar{z} + W \sim \mathcal{N}(0, 1/n + \rho^2)$. Independence with z^* gives the stated conditional distribution.

Under $b = 1$: $z^* = z_1$. Write $\bar{z} = z^*/n + (1/n)\sum_{i=2}^{n} z_i$. The sum on the right is independent of z^* with distribution $\mathcal{N}(0, n-1)$, so $(1/n)\sum_{i\ge 2} z_i \sim \mathcal{N}(0, (n-1)/n^2)$. Therefore

$$\theta \mid z^* \sim \mathcal{N}\left(\frac{z^*}{n},\ \frac{n-1}{n^2} + \rho^2\right).$$

(b) Let $V_1 = (n-1)/n^2 + \rho^2$ and $V_0 = 1/n + \rho^2$. Note $V_1 < V_0$ because $(n-1)/n^2 = 1/n - 1/n^2 < 1/n$. Then

$$\log \Lambda(\theta, z^*) = -\frac{(\theta - z^*/n)^2}{2V_1} + \frac{\theta^2}{2V_0} + \frac{1}{2}\log\frac{V_0}{V_1}.$$

Expanding,

$$\log \Lambda = \left(\frac{1}{2V_0} - \frac{1}{2V_1}\right)\theta^2 + \frac{z^*}{nV_1}\theta + \mathrm{const}(z^*).$$

The coefficient of θ^2 is negative (since $V_1 < V_0$ gives $1/V_0 < 1/V_1$), so $\log \Lambda$ is a concave (downward) parabola in θ. By Problem 5(b), the "declare member" region is an interval around the maximum.

(c) Under one-element neighbors with $|z_i|, |z_i'| \le 1$, the L_2-sensitivity of $\bar{z}$ is $|z_i - z_i'|/n \le 2/n$. The Gaussian-mechanism guarantee gives (ε, δ)-DP for

$$\rho \ge \frac{2}{n\varepsilon}\sqrt{2\log(1.25/\delta)}.$$

By Problem 2(b), the optimal MIA advantage is bounded by $\mathrm{Adv}^\star \le e^\varepsilon - 1 + \delta$. For $\varepsilon = 0.1, \delta = 10^{-5}$: $\mathrm{Adv}^\star \le e^{0.1} - 1 + 10^{-5} \approx 0.1052$, so the optimal MIA can correctly distinguish member from non-member with advantage at most $\approx 10.5\%$ above chance.

For $\varepsilon = 1$, the bound is $e - 1 \approx 1.718 > 1$, which is vacuous since the advantage is automatically in $[0, 1]$. Past $\varepsilon \approx \log 2 \approx 0.693$, the loose bound says nothing; the tighter Kairouz form is more informative.

7. **Influence-Function Unlearning (Closed Form).** Let $L(\theta; D) = \frac{1}{n}\sum_{i=1}^{n} \ell(\theta; z_i)$ be a twice-differentiable, λ-strongly convex loss in $\theta \in \mathbb{R}^d$. Recall that λ-strong convexity means the Hessian $H_\theta = \nabla^2 L(\theta; D)$ satisfies $H_\theta \succeq \lambda I$, which in turn implies H_θ is invertible with $\|H_\theta^{-1}\|_{\mathrm{op}} \le 1/\lambda$. Let $\theta_o = \arg\min_\theta L(\theta; D)$ and $\theta_{-z} = \arg\min_\theta L(\theta; D \setminus \{z\})$.

   (a) Consider the parametric family

   $$L_t(\theta) = L(\theta; D) - \frac{t}{n}\,\ell(\theta; z), \qquad t \in [0,1],$$

   and let $\theta(t) = \arg\min_\theta L_t(\theta)$. (Note that $L_0 = L(\cdot; D)$ gives $\theta(0) = \theta_o$, and L_1 is a positive scalar multiple of $L(\cdot; D \setminus \{z\})$, so $\theta(1) = \theta_{-z}$.) Use the first-order optimality condition $\nabla_\theta L_t(\theta(t)) = 0$ and differentiate with respect to t (the *implicit function theorem*) to derive

   $$\left.\frac{d\theta(t)}{dt}\right|_{t=0} = \frac{1}{n}\,H_{\theta_o}^{-1}\,\nabla_\theta \ell(\theta_o; z).$$

   (b) Conclude the first-order Newton-step approximation

   $$\theta_{-z} - \theta_o \approx \frac{1}{n}\,H_{\theta_o}^{-1}\,\nabla_\theta \ell(\theta_o; z).$$

   The "unlearned" parameter is then $\theta_u(D, z) = \theta_o + \frac{1}{n} H_{\theta_o}^{-1} \nabla_\theta \ell(\theta_o; z)$.

   (c) Assume in addition that ℓ is G-Lipschitz in θ (so $\|\nabla_\theta \ell(\theta; z)\| \le G$ for all θ, z). Show that the L_2-sensitivity of the unlearned parameter to the choice of deleted point is bounded:

   $$\sup_{z,z'} \|\theta_u(D, z) - \theta_u(D, z')\| \le \frac{2G}{\lambda n}.$$

   (Treat $H_{\theta_o}^{-1}$ as fixed; its operator norm is $\le 1/\lambda$ by strong convexity. This is the first-order sensitivity used in certified unlearning.)

Solution:

(a) **Implicit derivative.** By strong convexity the unique minimizer $\theta(t) = \arg\min L_t$ exists. The first-order condition is

$$F(\theta(t), t) := \nabla_\theta L(\theta(t); D) - \frac{t}{n}\,\nabla_\theta \ell(\theta(t); z) = 0.$$

Differentiate both sides with respect to t (chain rule):

$$\left[\nabla^2 L(\theta(t); D) - \frac{t}{n}\nabla^2 \ell(\theta(t); z)\right] \theta'(t) - \frac{1}{n}\nabla_\theta \ell(\theta(t); z) = 0.$$

At $t = 0$, the bracket equals $\nabla^2 L(\theta_o; D) = H_{\theta_o}$, and $\theta(0) = \theta_o$, so

$$\theta'(0) = \frac{1}{n}\,H_{\theta_o}^{-1}\,\nabla_\theta \ell(\theta_o; z).$$

(Invertibility of H_{θ_o} follows from $H_{\theta_o} \succeq \lambda I$.)

(b) **Taylor approximation.** A first-order Taylor expansion of $\theta(t)$ around $t = 0$ at $t = 1$ gives

$$\theta_{-z} - \theta_o = \theta(1) - \theta(0) \approx \theta'(0) \cdot 1 = \frac{1}{n}\, H_{\theta_o}^{-1}\, \nabla_\theta \ell(\theta_o; z),$$

so $\theta_u(D, z) = \theta_o + \frac{1}{n} H_{\theta_o}^{-1} \nabla_\theta \ell(\theta_o; z)$ is the Newton-step formula from the slides.

(c) **Sensitivity.** For two candidate deleted points z, z':

$$\theta_u(D, z) - \theta_u(D, z') = \frac{1}{n}\, H_{\theta_o}^{-1} (\nabla_\theta \ell(\theta_o; z) - \nabla_\theta \ell(\theta_o; z'))\,.$$

Take norms, using $\|H_{\theta_o}^{-1}\|_{\mathrm{op}} \le 1/\lambda$ and $\|\nabla_\theta \ell\| \le G$:

$$\|\theta_u(D, z) - \theta_u(D, z')\| \le \frac{1}{\lambda n}(G + G) = \frac{2G}{\lambda n}.$$

This is the L_2-sensitivity used to calibrate Gaussian noise in certified unlearning.

8. **Certified (ε, δ)-Unlearning by Gaussian Noise.** Building on Problem 7, let $A(D) = \arg\min_\theta L(\theta; D)$ be the (deterministic) full-retrain map, and define the noisy first-order unlearning algorithm

$$\mathcal{U}(D, z) = \theta_o(D) + \frac{1}{n}\, H_{\theta_o(D)}^{-1}\, \nabla_\theta \ell(\theta_o(D); z) + N, \qquad N \sim \mathcal{N}(0, \sigma^2 I_d).$$

Assume ℓ is λ-strongly convex and G-Lipschitz, and that the first-order approximation in Problem 7(b) has L_2 error bounded by

$$\eta := \|\theta_u^{\mathrm{1st}} - A(D \setminus \{z\})\| \le \frac{C_2\, G^2}{\lambda^2 n^2}$$

for some universal constant C_2 (a known second-order bound that you may use without proof).

(a) Show that the deterministic part of $\mathcal{U}(D, z)$ differs from $A(D \setminus \{z\})$ in L_2 norm by at most

$$\eta \le \frac{C_2\, G^2}{\lambda^2 n^2}.$$

(b) Recall the Gaussian-mechanism privacy bound: if $f, g : \mathcal{X}^n \to \mathbb{R}^d$ satisfy $\|f(X) - g(X)\| \le \eta$ for every X, then $f(X) + N$ and $g(X) + N$ are (ε, δ)-indistinguishable provided

$$\sigma \ge \frac{\eta}{\varepsilon}\sqrt{2 \log(1.25/\delta)}.$$

Apply this with $f = \theta_u^{\mathrm{1st}}$ and $g = A(\cdot \setminus \{z\})$ to conclude that

$$\sigma \ge \frac{C_2\, G^2}{\lambda^2 n^2\, \varepsilon}\sqrt{2 \log(1.25/\delta)}$$

yields an (ε, δ)-certified unlearning algorithm.

(c) How does the required noise scale with n? Compare to the Gaussian-mechanism noise needed for (ε, δ)-DP of the trained model itself ($\sigma \propto 1/(n\varepsilon)$). Explain in one or two sentences why the unlearning noise is smaller for large n.

Solution:

(a) **Sensitivity.** Define $\theta_u^{\mathrm{1st}} = \theta_o + \frac{1}{n} H_{\theta_o}^{-1} \nabla \ell(\theta_o; z)$, so $\mathcal{U}(D, z) = \theta_u^{\mathrm{1st}} + N$. The exact retrain is $\theta_{-z} = A(D \setminus \{z\})$. The hypothesis says

$$\|\theta_u^{\mathrm{1st}} - \theta_{-z}\| \le \eta \le \frac{C_2 G^2}{\lambda^2 n^2}.$$

So the deterministic part of $\mathcal{U}$ differs from $A(D \setminus \{z\})$ by at most η in L_2.

(b) **Gaussian-mechanism bound.** Apply the cited Gaussian-mechanism guarantee with $f = \theta_u^{\mathrm{1st}}$ and $g = \theta_{-z}$, both viewed as functions of the input (D, z):

$$\sigma \ge \frac{\eta}{\varepsilon}\sqrt{2\log(1.25/\delta)} \quad \Longleftarrow \quad \sigma \ge \frac{C_2 G^2}{\lambda^2 n^2 \varepsilon}\sqrt{2\log(1.25/\delta)}.$$

This matches the Guo et al. (2020) bound.

(c) Unlearning noise scales as $\sigma \propto 1/n^2$, whereas DP-of-the-trained-model noise scales as $\sigma_{\mathrm{DP}} \propto 1/n$. The unlearning task is *easier* for large n: a single training point's influence on the optimum is itself $O(1/n)$, and the Newton correction in Problem 7 reduces the residual to $O(1/n^2)$ — much smaller than what full DP would need to mask.

9. **NPO Has a Bounded Gradient; GA Does Not.** For LLM unlearning, let $\pi_\theta(y \mid x)$ denote the model's likelihood of completion y given prompt x, and let π_{ref} be a fixed reference model. The forget set is $\mathcal{D}_F$. Two unlearning losses:

$$\mathcal{L}_{\mathrm{GA}}(\theta) = -\,\mathbb{E}_{(x,y)\sim\mathcal{D}_F}[-\log \pi_\theta(y \mid x)] = \mathbb{E}[\log \pi_\theta(y \mid x)],$$
$$\mathcal{L}_{\mathrm{NPO}}(\theta) = \frac{2}{\beta}\,\mathbb{E}\left[\log\left(1 + \left(\frac{\pi_\theta(y \mid x)}{\pi_{\mathrm{ref}}(y \mid x)}\right)^{\beta}\right)\right] \quad (\beta > 0).$$

(a) Compute $\nabla_\theta \mathcal{L}_{\mathrm{GA}}$. Using the identity $\nabla_\theta \log \pi_\theta = \nabla_\theta \pi_\theta / \pi_\theta$, show that for any sample with $\pi_\theta(y \mid x) \to 0$ (and $\|\nabla_\theta \pi_\theta\|$ bounded away from 0 in some direction), $\|\nabla_\theta \mathcal{L}_{\mathrm{GA}}\| \to \infty$. Explain in one sentence why this drives the GA optimization to escape the data distribution ("GA collapse").

(b) Let $u(\theta) = \beta \log(\pi_\theta(y \mid x)/\pi_{\mathrm{ref}}(y \mid x))$. Show that

$$\mathcal{L}_{\mathrm{NPO}}(\theta) = \frac{2}{\beta}\,\mathbb{E}[\log(1 + e^{u(\theta)})], \qquad \frac{\partial \mathcal{L}_{\mathrm{NPO}}}{\partial u} = \frac{2}{\beta}\,\sigma(u) \in \left(0, \tfrac{2}{\beta}\right),$$

where σ is the logistic sigmoid. Conclude that

$$\|\nabla_\theta \mathcal{L}_{\mathrm{NPO}}\| \le 2\,\mathbb{E}[\|\nabla_\theta \log \pi_\theta(y \mid x)\|].$$

(c) Show that the right-hand bound above is uniform in θ: in particular, even as $\pi_\theta(y \mid x) \to 0$, the per-sample contribution $\sigma(u)\nabla_\theta \log \pi_\theta$ to the NPO gradient does *not* blow up, in sharp contrast to part (a). State briefly why this bounded-per-step divergence is what makes NPO "safe" for unlearning.

Solution:

(a) **GA gradient blowup.** By definition,

$$\nabla_\theta \mathcal{L}_{\mathrm{GA}}(\theta) = \mathbb{E}_{(x,y)\sim\mathcal{D}_F}[\nabla_\theta \log \pi_\theta(y \mid x)] = \mathbb{E}\left[\frac{\nabla_\theta \pi_\theta(y \mid x)}{\pi_\theta(y \mid x)}\right].$$

If a forget sample has $\pi_\theta(y \mid x) \to 0$ with $\|\nabla_\theta \pi_\theta(y \mid x)\|$ bounded away from 0 in some direction, the integrand blows up, so $\|\nabla_\theta \mathcal{L}_{\mathrm{GA}}\| \to \infty$. A single gradient-ascent step takes an arbitrarily large jump in θ, pushing the model off the data manifold and destroying fluency on *unrelated* inputs — the "GA collapse" observed by Jang et al.

(b) **NPO bounded gradient.** Let $u(\theta) = \beta \log(\pi_\theta(y \mid x)/\pi_{\mathrm{ref}}(y \mid x))$. Then $(\pi_\theta/\pi_{\mathrm{ref}})^\beta = e^{u(\theta)}$, so

$$\mathcal{L}_{\mathrm{NPO}}(\theta) = \frac{2}{\beta}\mathbb{E}[\log(1 + e^{u(\theta)})].$$

Differentiating $\log(1+e^u)$ with respect to u: $\frac{d}{du}\log(1+e^u) = e^u/(1+e^u) = \sigma(u) \in (0,1)$. So

$$\frac{\partial \mathcal{L}_{\mathrm{NPO}}}{\partial u} = \frac{2}{\beta}\sigma(u) \in \left(0, \frac{2}{\beta}\right).$$

By the chain rule, with $\nabla_\theta u = \beta\, \nabla_\theta \log \pi_\theta$,

$$\nabla_\theta \mathcal{L}_{\mathrm{NPO}} = \mathbb{E}\left[\frac{\partial \mathcal{L}_{\mathrm{NPO}}}{\partial u}\, \nabla_\theta u\right] = \mathbb{E}\left[\frac{2}{\beta}\sigma(u) \cdot \beta\, \nabla_\theta \log \pi_\theta\right] = 2\,\mathbb{E}[\sigma(u)\, \nabla_\theta \log \pi_\theta].$$

Since $\sigma(u) \in (0,1)$, taking norms gives $\|\nabla_\theta \mathcal{L}_{\mathrm{NPO}}\| \le 2\,\mathbb{E}[\|\nabla_\theta \log \pi_\theta\|]$.

(c) As unlearning progresses, $\pi_\theta(y \mid x) \to 0$ on forget samples, so $u \to -\infty$ and $\sigma(u) \to 0$. Concretely, for $u \to -\infty$, $\sigma(u) \sim e^u = (\pi_\theta/\pi_{\mathrm{ref}})^\beta$, so

$$\sigma(u)\, \nabla_\theta \log \pi_\theta \sim \frac{\pi_\theta^\beta}{\pi_{\mathrm{ref}}^\beta} \cdot \frac{\nabla_\theta \pi_\theta}{\pi_\theta} = \frac{\pi_\theta^{\beta-1}}{\pi_{\mathrm{ref}}^\beta}\, \nabla_\theta \pi_\theta,$$

which stays bounded for $\beta \ge 1$ (and remains tame in practice for smaller β). Operationally: once a forget sample's likelihood is small enough, NPO essentially stops updating θ on it. GA never stops — that is why it collapses. NPO's sigmoid saturation is the entire fix.

10. **Black-Box Unlearning Audits Require Many Queries.** Let θ_o and θ_u be two LLMs with output distributions P_0 and P_1 on a fixed prompt distribution. For two distributions P, Q on the same sample space, define the *total variation distance*

$$\mathrm{TV}(P,Q) = \sup_A |P(A) - Q(A)|,$$

where the supremum is over events A. Assume $\mathrm{TV}(P_0, P_1) \le \eta$. An auditor draws q i.i.d. prompts $x_1, \dots, x_q$, observes one sample $y_i \sim P_b(\cdot \mid x_i)$ per prompt, and outputs $\hat{b} \in \{0,1\}$. Write $P_b^{\otimes q}$ for the product distribution of the q observations under b.

(a) **(Le Cam two-point lower bound.)** Show that for any test based on q samples,

$$\Pr_{b=0}[\hat{b} \neq 0] + \Pr_{b=1}[\hat{b} \neq 1] \geq 1 - \mathrm{TV}(P_0^{\otimes q}, P_1^{\otimes q}).$$

Hint: Identify $\hat{b}$ with a function $\psi : \mathcal{X}^q \to [0,1]$ (the probability of declaring $b = 1$). Then prove the auxiliary fact: for any distributions P, Q and any $\psi : \mathcal{X}^q \to [0,1]$, $\mathbb{E}_P[\psi] - \mathbb{E}_Q[\psi] \leq \mathrm{TV}(P,Q)$.
(For the auxiliary fact, write $\psi(x) = \int_0^1 \mathbf{1}[\psi(x) > s]\, ds$ and integrate.)

(b) Show that $\mathrm{TV}(P_0^{\otimes q}, P_1^{\otimes q}) \leq q \cdot \mathrm{TV}(P_0, P_1) \leq q\eta$, by a hybrid (telescoping) argument: define $H_k = P_0^{\otimes k} \otimes P_1^{\otimes (q-k)}$ for $k = 0, \dots, q$, and bound $\mathrm{TV}(H_k, H_{k-1})$.

(c) Conclude that to guarantee total error $\leq 2\alpha$ (i.e., $\Pr_0[\hat{b} \neq 0] + \Pr_1[\hat{b} \neq 1] \leq 2\alpha$), the auditor needs

$$q \geq \frac{1 - 2\alpha}{\eta}.$$

Tighter χ^2-style bounds give $q = \Omega(1/\eta^2)$. Why does this make single-prompt audits of LLM unlearning fundamentally insufficient when θ_u, θ_o differ only by a small distributional perturbation?

Solution:

(a) **Auxiliary fact.** For any $\psi : \mathcal{X}^q \to [0,1]$, write $\psi(x) = \int_0^1 \mathbf{1}[\psi(x) > s]\, ds$ (since $\int_0^1 \mathbf{1}[\psi(x) > s] ds$ equals the length of $\{s \in [0,1] : s < \psi(x)\} = [0, \psi(x))$, which is $\psi(x)$). Integrating against any distribution R and swapping the order,

$$\mathbb{E}_R[\psi] = \int_0^1 R(\{\psi > s\})\, ds.$$

Therefore

$$\mathbb{E}_P[\psi] - \mathbb{E}_Q[\psi] = \int_0^1 [P(\{\psi > s\}) - Q(\{\psi > s\})]\, ds \leq \int_0^1 \mathrm{TV}(P,Q)\, ds = \mathrm{TV}(P,Q),$$

where the inequality uses $P(A) - Q(A) \leq \mathrm{TV}(P,Q)$ for every event A.
Le Cam. Identify $\hat{b}$ with the test $\psi(\cdot) = \Pr[\hat{b} = 1 \mid \cdot] \in [0,1]$. With $P = P_1^{\otimes q}$ and $Q = P_0^{\otimes q}$,

$$\Pr_0[\hat{b} \neq 0] + \Pr_1[\hat{b} \neq 1] = \mathbb{E}_Q[\psi] + \mathbb{E}_P[1 - \psi] = 1 - (\mathbb{E}_P[\psi] - \mathbb{E}_Q[\psi]) \geq 1 - \mathrm{TV}(P,Q).$$

(b) **Hybrid argument.** Set $H_k = P_0^{\otimes k} \otimes P_1^{\otimes (q-k)}$, so $H_q = P_0^{\otimes q}$ and $H_0 = P_1^{\otimes q}$. The TV distance satisfies the triangle inequality (apply $|P(A) - Q(A)| \leq |P(A) - R(A)| + |R(A) - Q(A)|$ and take supremum over A), so

$$\mathrm{TV}(P_0^{\otimes q}, P_1^{\otimes q}) = \mathrm{TV}(H_q, H_0) \leq \sum_{k=1}^{q} \mathrm{TV}(H_k, H_{k-1}).$$

Each pair H_k, H_{k-1} differs only in coordinate k: in H_k the k-th coordinate is drawn from P_0, in H_{k-1} from P_1. For any event $A \subseteq \mathcal{X}^q$, fix the other coordinates x_{-k} and let $A_{x_{-k}} = \{y : (x_1, \ldots, x_{k-1}, y, x_{k+1}, \ldots, x_q) \in A\}$ be the section. Then

$$H_k(A) - H_{k-1}(A) = \mathbb{E}_{x_{-k}}\left[P_0(A_{x_{-k}}) - P_1(A_{x_{-k}})\right],$$

and the inner difference is $\leq \mathrm{TV}(P_0, P_1)$ for every x_{-k}. Hence $H_k(A) - H_{k-1}(A) \leq \mathrm{TV}(P_0, P_1)$; taking the supremum over A gives $\mathrm{TV}(H_k, H_{k-1}) \leq \mathrm{TV}(P_0, P_1) \leq \eta$. Summing the telescope, $\mathrm{TV}(P_0^{\otimes q}, P_1^{\otimes q}) \leq q\eta$.

(c) Combining (a) and (b),

$$2\alpha \geq \Pr_0[\hat{b} \neq 0] + \Pr_1[\hat{b} \neq 1] \geq 1 - q\eta \implies q \geq \frac{1 - 2\alpha}{\eta}.$$

For $\alpha = 0.1$ this needs $q \geq 0.8/\eta$. Tighter Pinsker/Hellinger bounds with $\mathrm{TV}(P^{\otimes q}, Q^{\otimes q})^2 \leq 2q\,\mathrm{KL}(P\|Q)$ give the $q = \Omega(1/\eta^2)$ scaling (matching the χ^2 form behind Pawelczyk et al. 2024).

Implication for unlearning. If θ_u and θ_o differ in output distribution by a small TV η, then a black-box auditor with fewer than $\approx 1/\eta$ queries cannot reliably distinguish them. Single-prompt audits — the common regulatory-checkbox approach — fundamentally cannot certify that unlearning has happened; the auditor must use distributional probes (many queries) to detect the perturbation.